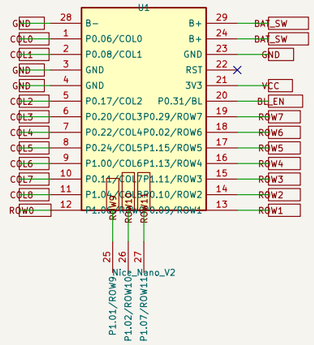


U1: user-installed TENSTAR / compatible nRF52840 module.
Module USB-C faces the rear case opening directly; no extension cable.
21 GPIO: 20 matrix + BL_EN. P0.09/P0.10 configured as GPIO.
No tall sockets; mechanical model defines mounting envelope.
Battery/charger and exact clone remain subject to bench validation.

Matrix

File: matrix.kicad_sch

Power

File: power.kicad_sch

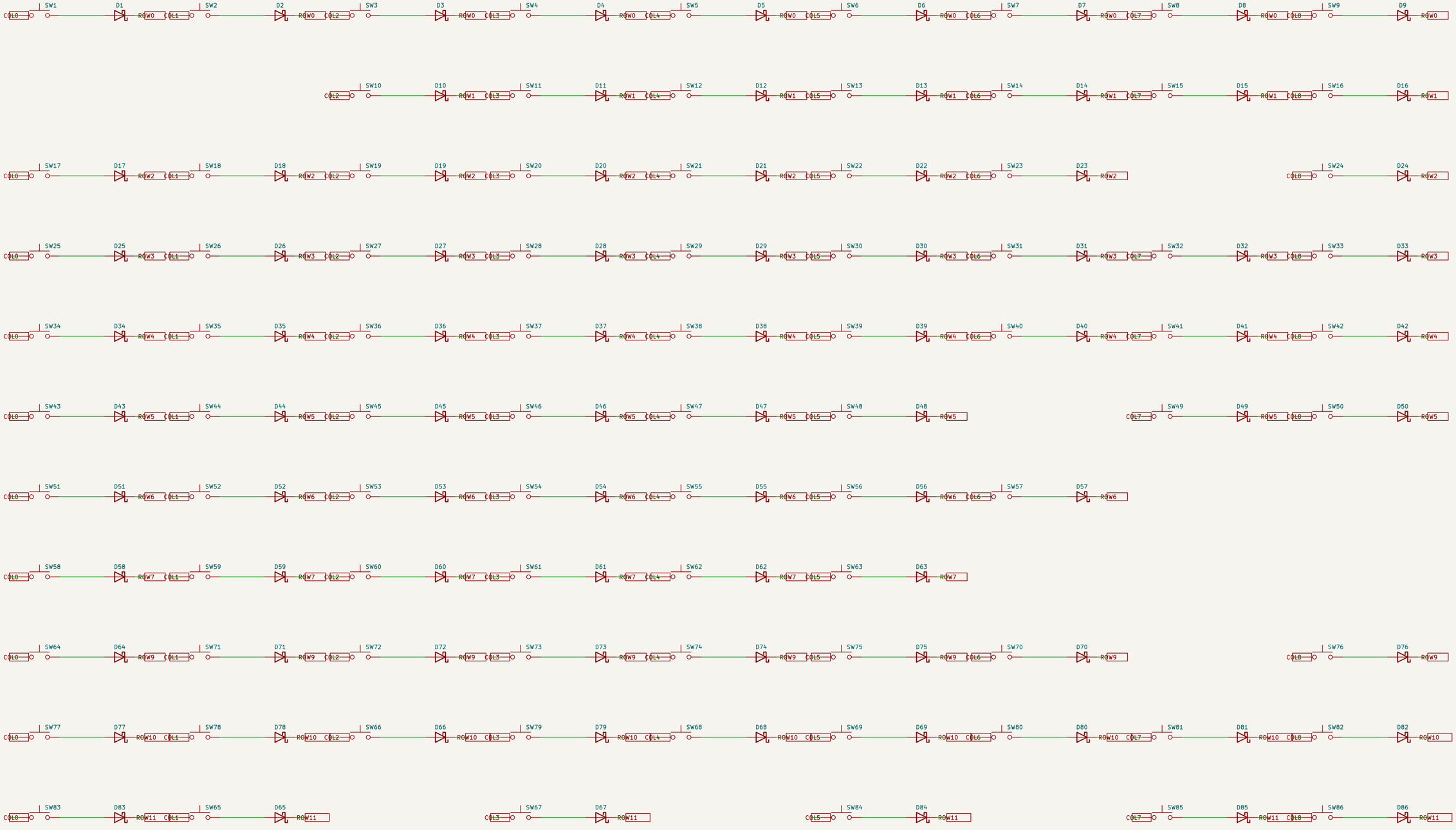
Pro Micro nRF52840 carrier; two layers; bottom assembly; module by user

Sheet: /
File: keyboard.kicad_sch

Title: TKL v12 — Controller

Size: A3	Date:	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 1/3

86 switches; SW9/SW15/SW16 restored; ROW8 unused. BAT46W diodes; module GPIO is 3.3 V.
V12: Choc V2 / Choc V1 / selected Gateron LP 3.0 solder patterns; alternate contacts share pins 1/2. See SWITCHES.md for mechanics.

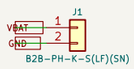


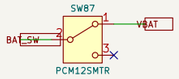
Pro Micro nRF52840 carrier; two layers; bottom assembly; module by user

Sheet: /Matrix/
File: matrix.kicad_sch

Title: TKL v12 – Matrix

Size: A3	Date:	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 2/3





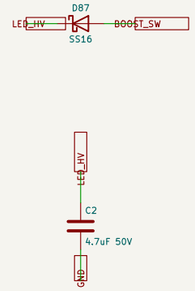


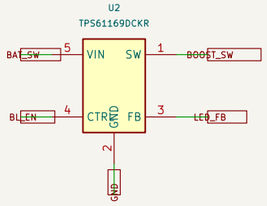


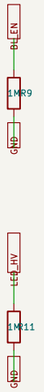


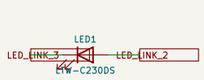


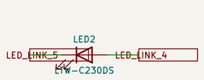


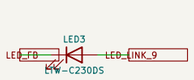


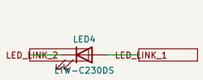


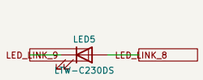


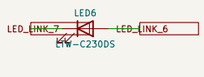


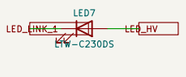


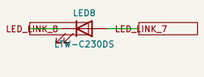


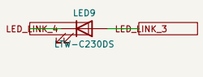


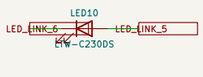












V12: ten bottom-mounted reverse LEDs shine through PCB into an acrylic plate.
U2 boost supply uses BAT_SW; one regulated SERIES string, not ten parallel branches.
R10 = 100 ohm: 204 mV / R10 = 2.04 mA nominal maximum at 100% PWM.
BL_EN is active-high 20 kHz PWM from existing module P0.31; low shuts down U2.
Remove LED7..LED10 and bridge EACH removed LED pads 1/2 for six-source mode.
LED1..LED6 remain distributed across both interior rows; see placement drawing.
Never bridge R10; do not bypass all LEDs. Power off before rework.
Protected 1S cell / module charging / RF and optical performance need bench tests.

Pro Micro nRF52840 carrier; two layers; bottom assembly; module by user

Sheet: /Power/
File: power.kicad_sch

Title: TKL v12 — Power

Size: A3	Date:	**Rev: 11**
KiCad E.D.A. 10.0.6		Id: 3/3